

Introduction to marketing mix modelling

International Marketing Analytics



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Where should the next euro go?

Alpenglow in 2025

6

markets: AT, DE, FR, IT,
NL, PL

5

channels: TV, online
video, paid search, paid
social, out-of-home

36 m

euros of media spend in
2025

1

Contribution: what did each channel
add to sales?

2

Return: which euro earned more, TV in
France or paid social in Poland?

3

Allocation: how should next year's fixed
budget be split?



The CEO's brief: the 2027 budget stays at the 2025 level. Move money only where the next euro earns more.

What marketing mix modelling is



Part 1 of 7 · definition, inputs and outputs,
MMM versus attribution and experiments,
history

Marketing mix modelling in one sentence



MMM explains weekly sales with marketing and non-marketing drivers, splits sales into a base and the contribution of each activity, and turns these into budget decisions.



Aggregate data

Weekly totals per country and channel
No tracking of individuals, so privacy-safe



A regression model

Sales explained by spend, price, season and more
Plus carryover and diminishing returns



Decisions

What each euro returned
Where the next euro should go

Little (1979): a model must capture

carryover

diminishing
returns

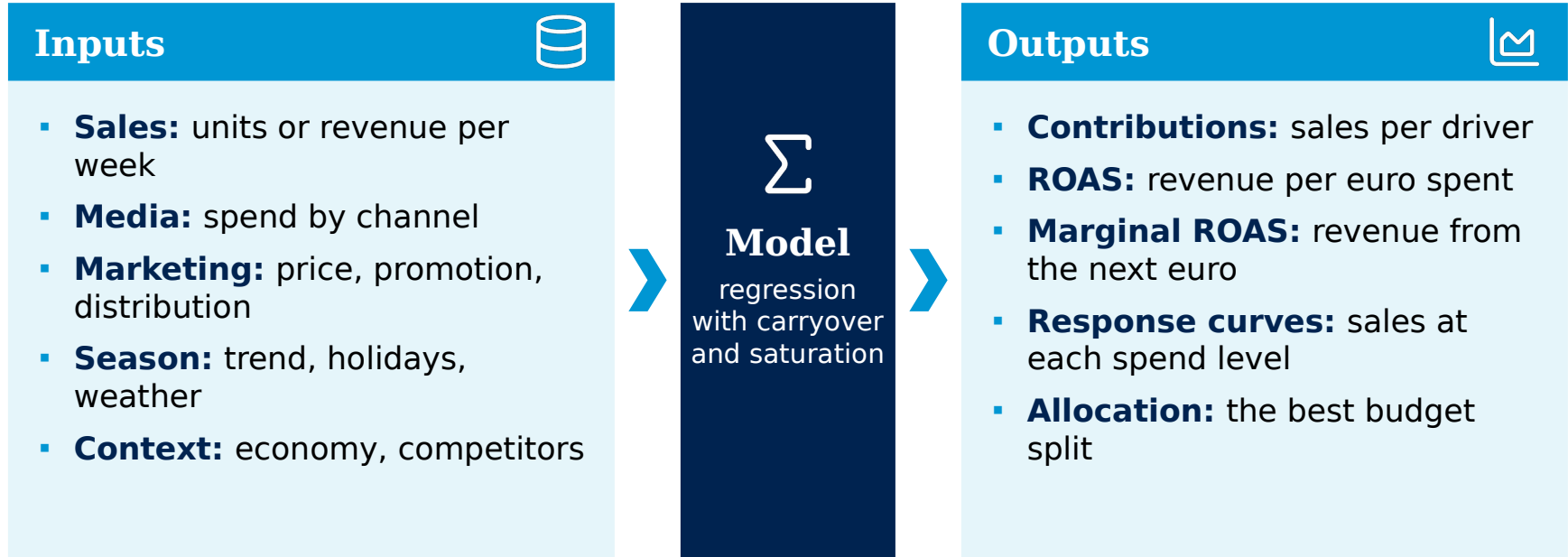
competition

interaction

dynamics

Sources: Little 1979; Hanssens, Parsons & Schultz 2001; Hanssens & Pauwels 2016

Inputs, model, outputs



In Alpenglöw: every input is a column of `data/mmm/alpenglöw_weekly.csv`

Three ways to measure marketing

	Marketing mix modelling	Multi-touch attribution	Experiments
Data	weekly totals per channel and country	clicks and views of individual users	test and control groups or regions
Answers	what each channel added, offline too	which touchpoints came before a sale	the causal lift of one change
Strength	all channels; privacy-safe	detailed and fast	most credible evidence of cause
Limit	correlation; needs variation in spend	digital only; credit is not cause	one channel, one period; costly



Triangulate: experiments calibrate the MMM, the MMM covers every channel, attribution steers digital campaigns day to day.

Sources: Google Modern Measurement Playbook 2023; IAB & MMA 2019; Gordon et al. 2019

An old method, popular again



Why now



- GDPR, cookie loss and Apple's tracking rules weaken user-level tracking
- MMM needs no personal data

64 %

of senior marketing leaders use MMM (Gartner 2024)

13 %

of advertisers turn it into insight quickly (WFA 2026)

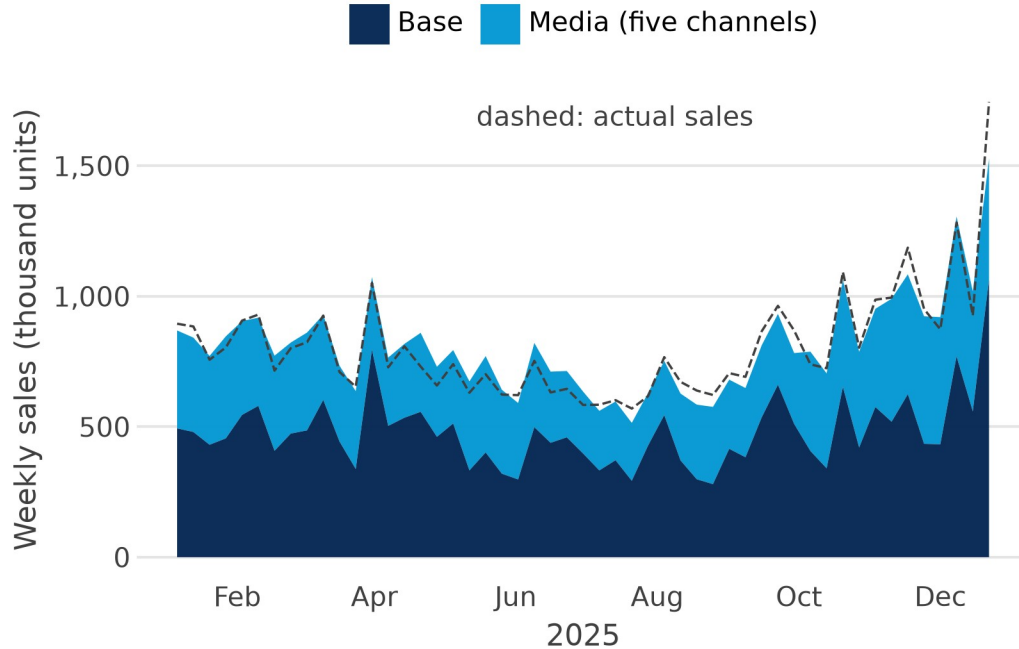
Sources: Clarke 1976; Little 1979; Hanssens et al. 2001; Jin et al. 2017; Gartner 2024; WFA & Ebiquity 2026

How MMM works



Part 2 of 7 · base and incremental sales, the equation, adstock, saturation, seasonality

Sales = base + incremental



Base: what would sell without media: price, distribution, season, trend

Incremental: what media add on top, channel by channel

Contribution: the sales that would disappear if a channel were switched off

Naive model: media $\approx 41\%$, too high

Alpenglow Germany, 2025, naive regression: budgets follow Christmas demand, so media get credit for seasonal sales (see the pitfalls slide).

Sources: Hanssens et al. 2001; Jin et al. 2017; own calculation on the Alpenglöck data

The regression behind it

Spend per
channel

Adstock:
carryover

Saturation:
less per €

× β :
the effect

Sales
contribution

Additive form

$$\text{sales}_t = \beta_0 + \sum \beta_c \cdot f(\text{spend}_c, t) + \gamma \cdot \text{controls}_t + \varepsilon_t$$

- β_c : extra units per unit of transformed spend
- **Good for:** contributions in units or euros

Log-log form

$$\ln \text{sales}_t = \beta_0 + \beta_p \cdot \ln \text{price}_t + \sum \beta_c \cdot \ln(1 + \text{adstock}_c, t) + \dots + \varepsilon_t$$

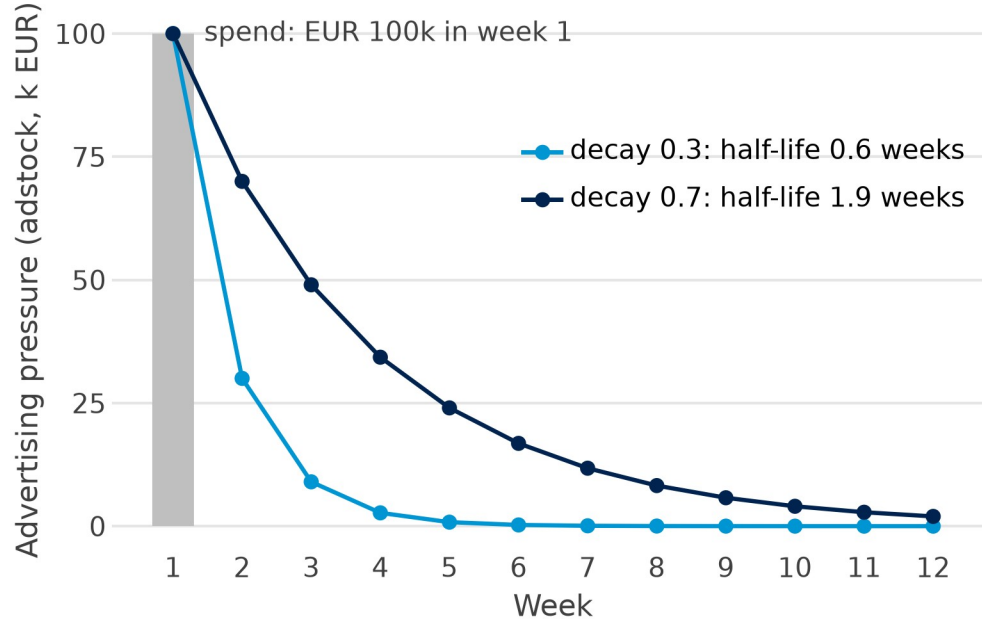
- β : elasticity, % change in sales per 1 % change
- **Good for:** comparing countries and drivers



Controls: price, promotion, distribution, trend, season, holidays, weather, economy, competitor spend.

Sources: Hanssens et al. 2001; Tellis 2006; Jin et al. 2017

Adstock: advertising keeps working



Geometric adstock

$$A_t = \text{spend}_t + \lambda \cdot A_{(t-1)}$$

λ (decay): share of last week's pressure that carries over

Half-life: $\ln 0.5 / \ln \lambda = 0.6$ or 1.9 weeks here

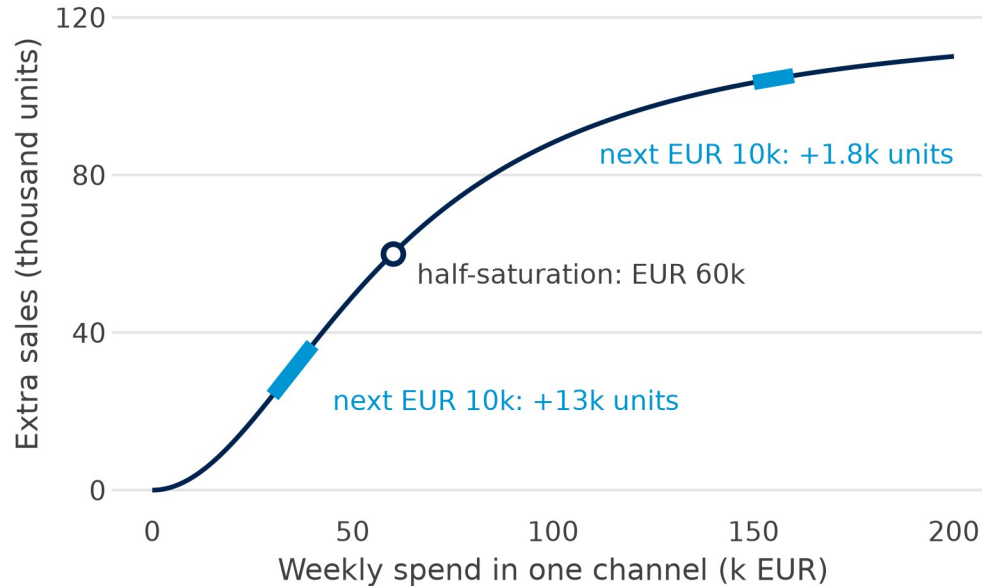
Total effect: $1/(1-\lambda) \times \text{week 1: } 1.4\times$ versus $3.3\times$



Without adstock, the weeks after a campaign are credited to whatever else happened then.

Sources: Little 1979; Jin et al. 2017; Robyn documentation (geometric and Weibull adstock)

Saturation: each extra euro buys less



Hill saturation

$$\text{Hill}(x) = x^s / (x^s + K^s)$$

K: spend at half the maximum effect

s: shape; above 1 gives an S-curve

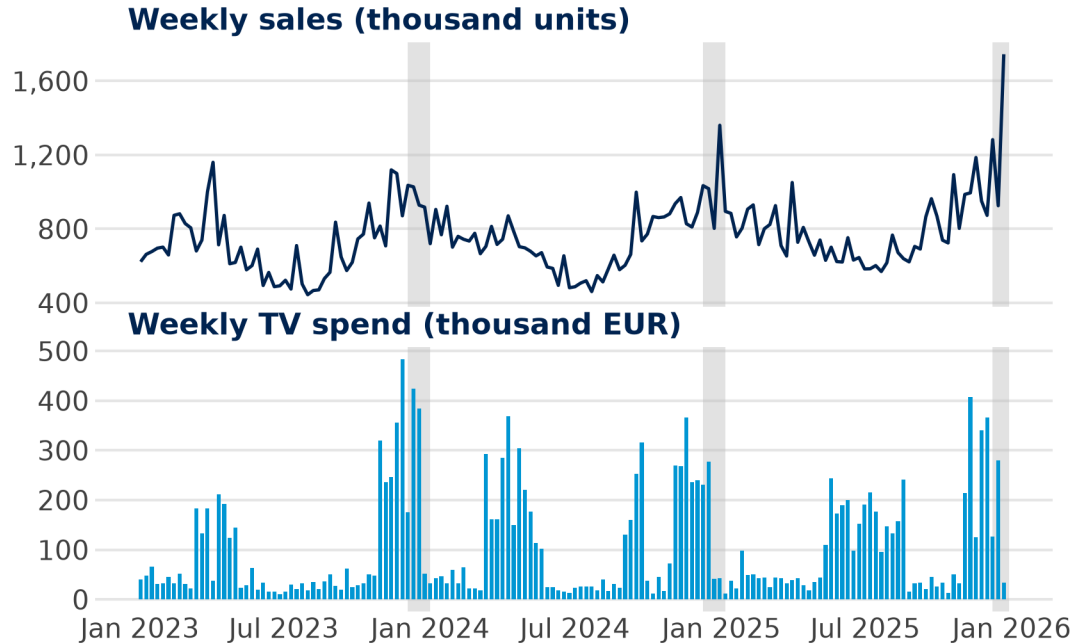
Marginal ROAS: revenue from the next euro, the slope of the curve



Decide at the margin: move money to where the next euro earns most.

Sources: Jin et al. 2017; Google Meridian documentation (media saturation and lagging)

Seasonality, trend and holidays



Alpenglow Germany, 2023–2025; grey: Christmas selling weeks.

Trend: slow growth or decline, a week counter

Seasonality: the yearly wave, Fourier terms

Holidays: dummies for Christmas, Easter, Valentine's

Weather, economy: temperature, consumer confidence



TV peaks when sales peak: does TV drive Christmas, or Christmas the TV budget?

Key terms (1): what the model splits



Marketing mix

The levers a brand controls: product, price, place (distribution) and promotion.

Alpenglow: price, promotions, distribution and five media channels



Base sales

What would sell without media: driven by price, distribution, season and trend.

The dark area in the decomposition chart



Incremental sales

What a channel adds on top of the base: the sales lost if it were switched off.

Also called the channel's contribution



Adstock

Advertising keeps working after the week it runs; the decay is the share that carries over.

Decay 0.7: half the effect is gone after about 2 weeks



Saturation

Each extra euro buys less: the response flattens as spend grows.

The next EUR 10k buys far fewer units at high spend



Control variables

Non-media drivers in the model, so media are not credited with their effect.

Price, promotions, holidays, temperature, competitors

Key terms (2): what the model tells you

Elasticity

The % change in sales for a 1 % change in a driver; the coefficient in a log-log model.

Price elasticity –1.4: 1 % higher price, 1.4 % fewer units

Response curve

Incremental sales at each spend level: adstock and saturation in one picture.

Read it to see where a channel stops paying off

ROAS and ROI

ROAS: incremental revenue per euro spent. ROI: incremental profit per euro spent.

ROAS 3: every euro returned EUR 3 of revenue on average

Marginal ROAS

The return on the next euro: the slope of the response curve.

Move budget to the highest marginal, not average, ROAS

Holdout

Fit on past weeks, predict weeks the model has not seen, compare the errors.

Fit 2023-2024, forecast 2025, report the MAPE

Calibration

Checking or anchoring model effects with experiments such as geo lift tests.

Paid-social test in 12 of 40 German regions

What we know: empirical generalisations



Part 3 of 7 · elasticities, carryover and wear-out, differences between countries

Elasticities: price versus advertising

Advertising: the meta-analyses

0.12 short-term elasticity, mean
(Sethuraman et al. 2011)

0.24 long-term elasticity, mean (same
study)

0.09 mean in an international
database (Henningesen et al.
2011)

$\approx 1/2$ of 389 TV weight tests: no
significant sales effect (Lodish et
al. 1995)

Alpenglow Germany: first look

-1.4 price: 1 % lower price, about
1.4 % more units

0.06 TV: 1 % more TV spend, about
0.06 % more units

Simple log-log regression, 2023-2025. You estimate
it properly, by country, in Session 2.



Reading 0.12: 10 % more advertising lifts sales by about 1.2 %.

How long effects last

60 %

of the effect came after week 13 (141 UK brands; Ebiquity & Thinkbox 2024)

2×

long-term elasticity versus short-term: 0.24 against 0.12 (Sethuraman et al. 2011)



Data interval matters

Monthly or annual data suggest longer carryover than weekly data (Clarke 1976; Köhler et al. 2017)



Wear-in and wear-out

Ads may need repetition to work, and lose effect when shown too often or too long (Tellis 2004; Vakratsas & Ambler 1999)



For your model: estimate carryover per channel from weekly data and check half-lives against these benchmarks.

Effects differ between countries



Market type: advertising matters more in emerging markets; price and distribution effects differ too (Bahadir et al. 2015, 14 markets)



Business cycle: in downturns price sensitivity rises and advertising works less (van Heerde et al. 2013)



Culture: how strongly ad budgets follow the economy depends on national culture (Deleersnyder et al. 2009, 37 countries)



Payoff: reallocating across countries with elasticities: about EUR 500m more profit at Bayer (Fischer et al. 2011)



For Alpenglow: expect Poland to respond differently from Germany. Hierarchical models let countries differ while borrowing strength from each other (Sun et al. 2017).

Doing it well



Part 4 of 7 · data, pitfalls, validation, Bayesian priors, open-source tools

What the data must offer



Two to three years, weekly

Seasons must repeat; Alpenglow has 156 weeks per country



Variation in spend

Channels must move up and down, and not always together



Fine geography

Countries or regions, not one national total (Nielsen 2022)



All drivers

Price, distribution, season and competitors, not only media

+68 %

A media-only model overstated advertising ROI by 68 % compared with a model that included the other drivers (Nielsen 2022).

Four pitfalls

Endogeneity



Budgets follow demand: German TV spend nearly doubles in Christmas weeks

Multicollinearity



Channels that move together cannot be told apart; check the VIF

Overfitting



Many parameters, few weeks: the fit looks great, the forecast fails

Identification



Saturation and effects that change over time can mimic each other

10.6 vs 4.2 **Paid-search ROAS:** a standard MMM against geo experiments at Zalando (Heusch 2026).

Sources: Chan & Perry 2017; Dew et al. 2024; Heusch 2026

Validate before you trust



Holdout 	Plausibility 	Experiments 
▪ Fit to 2023-2024 ▪ Forecast 2025 ▪ Compare errors (MAPE)	▪ Media effects positive ▪ Half-lives and elasticities near benchmarks ▪ Stable when data change	▪ Spend up or off in test regions ▪ Lift = test minus control ▪ Alpenglow: paid social, 12 of 40 German regions



Why experiments: across 15 and then 663 Facebook experiments, non-experimental estimates were often far from the true lift.

Sources: Vaver & Koehler 2011; Gordon et al. 2019; Gordon, Moakler & Zettelmeyer 2023

Bayesian MMM: start from what we know

Prior

what we believe first:
elasticities near 0.1, a lift-
test result



Data

156 weeks of sales, spend
and controls per country



Posterior

updated belief: every effect
and ROAS with a range

Why it suits MMM



- Short series, many parameters: priors keep estimates sensible
- Uncertainty for every output
- Experiments enter as priors

What to watch



- Weak data: the prior decides; compare prior and posterior
- Check that the sampler converged
- State and justify every prior

Sources: Jin et al. 2017; Hanssens 2018; Zhang et al. 2024 (Google)

Open-source MMM tools

	Robyn	Meridian	PyMC-Marketing
Maker	Meta	Google	PyMC Labs
Language	R (Python beta)	Python	Python
Method	ridge regression with automated search	hierarchical Bayesian, by geo	Bayesian, modular
Strengths	calibration, budget allocator, widely used	many geos, reach and frequency, ROI priors	lift-test calibration, time-varying effects, optimiser
Version	3.12.0 (Dec 2024)	2.1.0 (Sep 2026)	1.2.0 (Sep 2026)

◀ ▶ **In this course:** PyMC-Marketing and Meridian in Python, Session 5.

Sources: Runge et al. 2024; Meridian and PyMC-Marketing documentation; open-source MMM review (2026)

Ad spend and attribution



Part 5 of 7 · where the money goes, last click,
why attribution misleads, real-world tests

Where the advertising money goes

\$1.14 tn global advertising spend in 2025,
+8.8 %

73 % of it is digital: search, social, video,
retail media

€119 bn digital advertising in Europe in 2024,
+16 %

67 % of all ad spend in Europe is digital

What this means



- Digital ads are bought click by click
- The platforms that sell the ads also report their success
- Small measurement errors move billions



The question of this part: who gets the credit for a sale, and is it deserved?

Sources: WPP Media, This Year Next Year, mid-year 2025; IAB Europe, AdEx Benchmark 2024

Attribution: who gets the credit?



Credit for the order	TV	Instagram	YouTube	Search
Last click	0 %	0 %	0 %	100 %
First click	0 %	100 %	0 %	0 %
Linear	0 %	33 %	33 %	33 %
Data-driven (GA4)	0 %	e.g. 25 %	e.g. 15 %	e.g. 60 %



All four models share out one sale: none asks if it would have happened without ads.

Sources: Google Analytics 4 documentation; journey and data-driven shares are illustrative

Last click: simple, common and biased

How it works



- 100 % of the credit to the last ad clicked before the order
- Easy to compute, easy to explain
- GA4 and Google Ads: since 2023 only last click and data-driven are left

Who wins, who loses



- Wins: branded search, retargeting, coupon and affiliate sites
- Loses: TV, online video, social, posters
- Ignores: views without a click, offline sales

The catch: the last click often comes from people who had already decided to buy

Example: after a TV spot, a viewer searches “alpenglow”: search gets the sale, TV gets nothing

The result: budgets drift to the end of the funnel, and the demand that feeds it slowly dries up

Sources: Search Engine Land 2023 (Google Ads and GA4); Li & Kannan 2014 (Journal of Marketing Research)

Six problems of attribution models

Correlation only

Ads target people who buy anyway
Credit is not extra sales

Tracking gaps

Cookie consent, ad blockers, Apple's ATT
2021: 4 % of US iPhone users allowed tracking

Offline is invisible

TV, radio, posters, word of mouth
Sales in shops have no click

Double counting

Each platform counts its own orders
Sum of reports > real orders

Click fraud

Affiliates claim sales that come anyway
Uber: fake clicks on free installs

Many devices

Phone on the sofa, laptop at work
One buyer looks like three



The remedy: experiments (ads off in some regions) and MMM, which needs no tracking.

Sources: Gordon et al. 2019 (Marketing Science); Flurry Analytics 2021; WARC 2019 (Uber)

When the ads were switched off

99.5 %

eBay: brand-keyword traffic that came back through the free links when the ads were paused

\$100 m

Uber: of \$150 m yearly performance ads switched off, with no change in app installs

\$200 m

P&G: of digital ads cut in 2017; reach rose about 10 %

95 %

Airbnb: of web traffic kept in 2020 after cutting marketing by more than half

15

Facebook experiments: click-based methods often missed the true lift, mostly overstating it



Lesson: test what the ads add; do not trust the click report.

Sources: Blake, Nosko & Tadelis 2015; WARC 2019; Adweek 2018; Campaign Asia 2021; Gordon et al. 2019

Discuss: who sold the chocolate?

Pairs: 10 minutes

Alpenglow Germany, November

Google Ads (last click) **3,900**

Meta (click or view) **2,600**

Affiliate sites **1,300**

Sum of the reports **7,800**

Orders in the web shop **6,000**

Plenary: 5 minutes

- 1 Why** do the reports add up to 7,800 orders when the shop had 6,000?
- 2 Branded search** shows a last-click ROAS of 12, TV of 2. The CFO wants to move TV money to search. Your answer?
- 3 Design a test:** how would you find out what branded search really adds?
- 4 MMM:** what can it see that attribution cannot, and what can it not see?

Sources: Illustrative numbers; real-world cases on the previous slide

Reading: Analytics for Marketers



Part 6 of 7 · Fantini & Narayandas (HBR 2023)
and an in-class discussion

Three approaches to analytics

Descriptive: humans decide

Predictive: machines predict, humans choose

Prescriptive: machines decide

What happened?



- Dashboards of aggregated past data
- Managers add experience and gut feeling
- Low pain, low gain

What if ...?



- Regression, elasticities, forecasts
- Finer, tactical decisions
- Few variables, uncertain inputs

What to do?



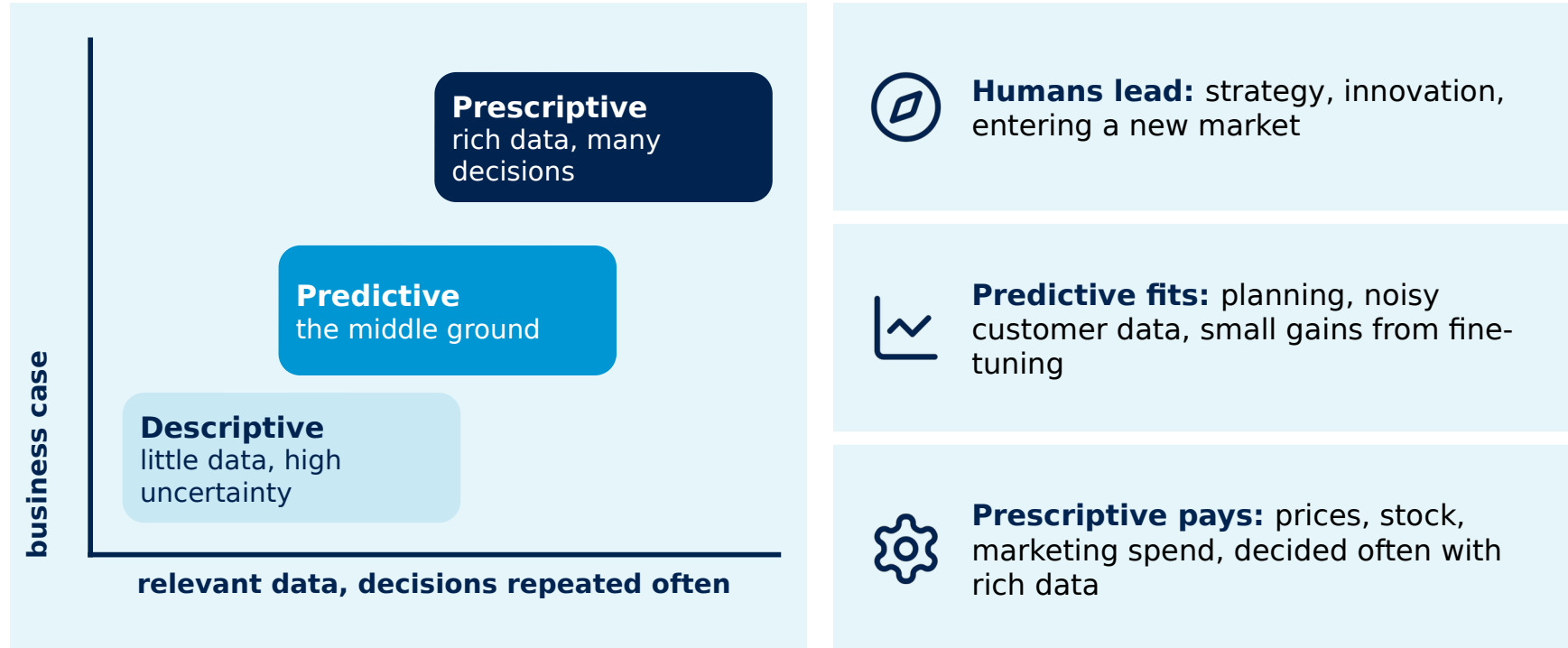
- Optimises towards goals and rules set by managers
- Learns from many cheap experiments
- Costly to build and run



MMM today: predictive. With a budget optimiser on top it becomes prescriptive.

Sources: Fantini & Narayandas 2023 (Harvard Business Review)

When to use which approach



Sources: Fantini & Narayandas 2023 (Harvard Business Review); simplified illustration

Case: markdowns at Event Network

1 Descriptive

- Discount SKUs with the most days of stock
- Until the markdown budget is spent
- Ignores customers and context

2 Predictive

- Regression of volume on price
- Elasticity by category, store and week
- Crude, but better than the rule

3 Prescriptive

- Machine learning plus optimisation
- Many data sources, 100,000+ SKUs
- Learns and improves over time

-2

price elasticity

+20 %

units after a 10 % price cut

+8 %

revenue: \$1,000 → \$1,080

10-20 %

of sales explained by price



Lesson: a crude model can still beat a rule of thumb.

Sources: Fantini & Narayandas 2023 (Harvard Business Review)

Accuracy is not the goal

Data scientists



- Aim: the most accurate model
- Judge by statistical fit

Business scientists



- Aim: the most profitable decision
- Judge by the cost of each error

False positive: predicted a win that does not come: wasted sales and marketing effort

False negative: predicted a loss that would have been a win: a lost opportunity

In this course: Session 3 and case task C7 set the churn threshold by profit, not accuracy



The manager's new role: “from the person who has all the answers to the one who asks the right questions”

Sources: Fantini & Narayandas 2023 (Harvard Business Review)

In-class discussion: six Alpenglow decisions

	Decision	How often	Data available
1	Split next year's media budget over 6 countries and 5 channels	once a year	3 years of weekly sales and spend
2	Set the bids for paid-search keywords	every day	clicks, costs and orders per keyword
3	Price of the Christmas gift box in Poland	once a year	156 weeks of prices and sales
4	Choose which Club members get a retention offer	every month	6,000 subscribers and their history
5	Enter Swiss supermarkets, beyond the web shop	once	almost none
6	Discount unsold Easter chocolate after Easter	every year, 200 SKUs × stores	sales and stock per SKU and store

Your task: for each decision, choose descriptive, predictive or prescriptive, and say who decides.

Discuss in your group

**Groups of 3-4:
15 minutes**

**Plenary:
10 minutes**

**Hand in: one
photo of your table**

- 1 Classify** the six decisions with the article's two criteria: relevant data and the business case
- 2 Alpenglöw's MMM:** predictive or prescriptive? Should a machine decide the yearly budget?
- 3 Crude models:** Event Network's price model explained 10-20 % of sales. When is that good enough?
- 4 Errors:** for the retention offer, which costs more: a false positive or a false negative?
- 5 The manager's job:** write one question for Alpenglöw's CMO that no model can answer

Sources: Reading: Fantini & Narayandas 2023, Analytics for Marketers, Harvard Business Review (on Canvas)

MMM in this course



Part 7 of 7 · how the sessions build the skills,
the group case study

Five sessions, one model

1	Introduction and linear regression	OLS, model fit and interpretation	→ the base regression
2	Advanced regression	log-log elasticities, dummies, non-linear effects, moderation and mediation, adstock and saturation, diagnostics	→ the MMM itself
3	Logistic regression	binary outcomes: purchase, churn; odds ratios	→ customer-level questions
4	ARIMA	trend, seasonality, ARIMA and ARIMAX forecasts	→ the sales baseline
5	Advanced MMM tools	Bayesian MMM, budget optimisation	→ the full workflow

The group case study

**S1: what drives
weekly orders?**

**S2: price and
advertising**

**S3: who will
cancel?**

**S4: Q1 2026
baseline**

The brief



- Alpenglöck's web shop in AT, DE and CH
- Four questions from the Head of E-commerce

The data



- Weekly shop data, 3 countries × 156 weeks
- Club subscribers
- The Q1 2026 plan

You deliver



- A Quarto report with all code
- Groups of at most 4
- 30 % of the grade



Every number from code: the report renders from a fresh copy of your project folder.

Brief, data and assessment criteria: assignments/group-project/ and Canvas

Key takeaways

- 1 **MMM** explains weekly sales with marketing and other drivers and splits them into base and incremental
- 2 **Adstock and saturation** capture carryover and diminishing returns; decide on marginal, not average, ROAS
- 3 **Benchmarks:** advertising elasticities around 0.1 short term, larger price effects, differences by country
- 4 **Budgets follow demand:** watch endogeneity; validate on a holdout and calibrate with experiments
- 5 **Open-source Bayesian tools** make it doable in Python: you will build one



The question stays the same: where should the next euro go?

Foundations and evidence



Hanssens, D. M., Parsons, L. J. & Schultz, R. L. (2001). Market Response Models. 2nd ed. Kluwer.

Sethuraman, R., Tellis, G. J. & Briesch, R. A. (2011). How well does advertising work? Journal of Marketing Research 48(3), 457–471.

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Modern MMM and tools



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github.com/google/meridian

PyMC Labs. PyMC-Marketing. Documentation and code (Python).

github.com/pymc-labs/pymc-marketing



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